

Vertex excess and the sharp exponent in simplex stability

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Abstract

For a full-dimensional convex d -polytope with at most $d + 1 + r$ vertices, we determine the sharp power of the projection-body pyramid defect that controls containment in a dilation of every prescribed maximum-volume inscribed simplex, centered at that same simplex's centroid. The exponent is $1/\min(r, d - 1)$. We give an explicit all-scale upper bound and a truncation-and-pyramid family with an exact limiting lower constant. The geometric step is a cone-volume estimate controlled by the number of inner apex points: collapsing two base rays under projection improves the usual cap exponent. A finite law on the actual facets supplies the facet incidence needed to convert a vertex bound into this apex count. We include the finite determinant assignment, enclosure, and prescribed-simplex retention arguments so that the dependence on the original geometric defect can be checked directly.

Verification status. The underlying traditional proof and sharpness construction received a separate analytic review by an independent model, with a PASS disposition. This is not professional peer review. The present manuscript is an expanded exposition of that reviewed argument; the new vertex-excess theorem has not been formalized in Lean. Earlier Lean results are identified in Appendix A, with no claim that an older compiled main theorem certifies the new result. No priority or first-discovery claim is made.

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1 The invariant and the main theorem

For a full-dimensional convex body $K \subset \mathbb{R}^d$, let $|K|$ denote its Euclidean volume. For a *unit* vector u , let $\pi_K(u)$ be the $(d-1)$ -dimensional volume of its orthogonal projection onto $u^\perp$. We reserve the homogeneous notation

$$b_K(z) = \begin{cases} \|z\| \pi_K(z/\|z\|), & z \neq 0, \\ 0, & z = 0 \end{cases}$$

for nonunit vectors. The projection body ΠK has support function $h_{\Pi K} = b_K$. If L has dimension p , set

$$R(L) = \frac{|\Pi L|}{|L|^{p-1}}, \quad \text{pyr } L = \text{conv}((L, 0) \cup \{(0, 1)\}).$$

We use exactly the projection-body pyramid invariant of the retained Entry 005 project:

$$a(K) = \left(\frac{d}{d+1}\right)^d \frac{R(\text{pyr } K)}{R(K)} - 1, \quad e(K) = a(K) - \frac{1}{d+1}. \quad (1.1)$$

For a full-dimensional simplex $S \subset K$ with centroid g_S , define

$$E(K, S) = \inf\{\tau \geq 0 : K \subset g_S + (1 + \tau)(S - g_S)\}. \quad (1.2)$$

The dilation factor itself is $1 + E$. The simplex in this definition is prescribed; neither the simplex nor its center is optimized in $E(K, S)$.

Theorem 1.1 (Sharp vertex-excess exponent). *Let $d \geq 3$ and $r \geq 1$ be integers, and put*

$$n = d + 1, \quad k = \min(r, d - 1), \quad H_d = 3d^2(d + 1)^3(d + 2), \\ T_{d,r} = \min\{64(d + 1)r, 16(d + 1)^2\}.$$

For every full-dimensional convex d -polytope K with at most $d + 1 + r$ vertices, and every prescribed maximum-volume inscribed simplex S ,

$$E(K, S) \leq T_{d,r} H_d^{1/k} e(K)^{1/k}. \quad (1.3)$$

The invariant $e(K)$ is nonnegative. The exponent $1/k$ is optimal: there is an explicit family (K_t, S_t) in this same class, with S_t maximum and $e(K_t) \rightarrow 0$ as $t \downarrow 0$, such that

$$\lim_{t \downarrow 0} \frac{E(K_t, S_t)}{e(K_t)^{1/k}} = n \left(\frac{n^2}{k(k+1)} \right)^{1/k}. \quad (1.4)$$

Consequently no exponent $\beta > 1/k$ admits a uniform bound near zero defect, even with an arbitrary finite coefficient depending on d, r .

One extra vertex gives a linear defect estimate. Two extra vertices give exponent $1/2$ in every dimension $d \geq 3$. Saturation at the unrestricted exponent $1/(d-1)$ occurs when the vertex allowance reaches $2d$. The coefficient in (1.3) is explicit but is not asserted to be optimal. The case $r = 0$ is trivial: a full-dimensional polytope with at most $d + 1$ vertices is a simplex. The theorem concerns $d \geq 3$; the two-dimensional body used in the sharpness construction is computed explicitly below.

The proof separates three geometric operations. First, a finite apex count controls the volume missing from a cone and survives a ray-collapsing projection. Second, actual facet normals yield an enclosing simplex with a relative projection error of at most $H_d e(K)$. Third, a near-vertex condition for that enclosing simplex controls the originally prescribed maximum, including maxima whose vertices are not vertices of K .

1.1 Context and scope

Mixed-volume polynomials, their monotonicity, the first mixed-volume formula, and Minkowski's first inequality are classical inputs [2, 1]. They are not new ingredients of this work. Simplex stability for other functionals is studied, for example, by Schneider [3]; cone-volume stability for subspace concentration and the U -functional is treated in [4]. Projection-body volume questions also have substantial earlier literature [6, 5]. These neighboring statements involve different functionals or classes and are not used to infer a novelty claim here. The source comparison accompanying this manuscript is focused rather than exhaustive; in particular it does not certify the complete overlap with all earlier projection-body results.

The cone count, projection collapse, and actual-facet specialization are the added steps relative to the retained unrestricted proof. For transparency, we also reproduce finite versions of its determinant assignment and retention arguments. Thus the mathematical chain below can be read without running the earlier formal source tree.

2 Cone volume and a ray-collapsing projection

Lemma 2.1 (Finite apex count). *Let N, m be positive integers. Let B be a compact convex $(N-1)$ -body in the affine hyperplane $\ell = 1$ of $\mathbb{R}^N$, where ℓ is linear, and suppose $C = \text{conv}(0, B)$ is full dimensional. For $z_1, \dots, z_m \in B$ and $0 \leq \rho \leq s_j \leq 1$ with $\rho < 1$, set*

$$L = \text{conv}(B, s_1 z_1, \dots, s_m z_m).$$

Then

$$\frac{|L|}{|C|} \leq 1 - \rho^{\min(m, N)}. \quad (2.1)$$

The exponent and constant are sharp.

Proof. If $\rho = 0$, the assertion is $L \subset C$. For $N = 1$, the base is a point and the interval length ratio is at most $1 - \rho$. We may thus suppose $N \geq 2$ and $0 < \rho < 1$. Since

$$s_j z_j = \frac{1 - s_j}{1 - \rho} \rho z_j + \frac{s_j - \rho}{1 - \rho} z_j,$$

we have $L \subset \text{conv}(B, \rho Z)$, where $Z = \text{conv}(z_1, \dots, z_m)$ and $h = \dim Z \leq \min(m - 1, N - 1)$. Put $q = N - 1$. At height $\ell = s \in [\rho, 1]$, the latter hull has section

$$aB + bZ, \quad a = \frac{s - \rho}{1 - \rho}, \quad b = \frac{\rho(1 - s)}{1 - \rho}. \quad (2.2)$$

To interpret its q -volume, choose $b_0 \in B$ and subtract sb_0 . Since $a + b = s$, the translated section is $a(B - b_0) + b(Z - b_0)$ in $\ker \ell$.

Normalize mixed volume by $V(B[q]) = |B|_q$. The classical volume polynomial is

$$|aB + bZ|_q = \sum_{j=0}^q \binom{q}{j} a^{q-j} b^j V(B[q-j], Z[j]). \quad (2.3)$$

Translation invariance and monotonicity give $0 \leq V(B[q-j], Z[j]) \leq |B|_q$. For $j > h$ these terms vanish: translate Z into an h -dimensional linear subspace; a product box bounds $|B + tZ|_q$ by $O(t^h)$ for $t \geq 1$, and nonnegativity of the polynomial coefficients forces every higher coefficient to vanish. This also follows from the essential-collection criterion in [1, Theorem 2.2].

The substitution $s = \rho + (1 - \rho)v$ gives

$$\binom{q}{j} \int_{\rho}^1 a^{q-j} b^j ds = (1 - \rho) \rho^j \binom{q}{j} \int_0^1 v^{q-j} (1 - v)^j dv = \frac{(1 - \rho) \rho^j}{q + 1}.$$

The normal-height Jacobian is $1/\|\ell\|$, while $|C| = |B|_q/(N\|\ell\|)$. Consequently

$$\frac{|L|}{|C|} \leq (1 - \rho) \sum_{j=0}^h \rho^j = 1 - \rho^{h+1} \leq 1 - \rho^{\min(m, N)}.$$

For sharpness when $m \leq N$, take $C = \text{conv}(0, e_1, \dots, e_N)$, $B = \text{conv}(e_1, \dots, e_N)$, and $z_j = e_j$. The omitted region has the volume of the simplex with intercepts ρ on the first m coordinate rays and one on the others, namely $\rho^m |C|$. For $m > N$, include all N base vertices and add redundant points; the omitted volume is $\rho^N |C|$. $\square$

Remark 2.2 (Factorial convention). With the diagonal normalization above, the coefficient of $\lambda_1 \cdots \lambda_q$ in $|\lambda_1 K_1 + \cdots + \lambda_q K_q|_q$ is $q!V(K_1, \dots, K_q)$. Grouping repeated factors gives exactly the binomial coefficient in (2.3). The coefficient statement in [1, Theorem 2.1, version 2] appears to omit this factorial relative to that paper's definition (1.1). We use its classical background and multilinearity with the normalization stated here, not that coefficient statement literally. At $q = 2$ and $K_1 = K_2 = B$, the cross coefficient of $(\lambda_1 + \lambda_2)^2 |B|_2$ is already $2|B|_2$.

Lemma 2.3 (Projection collapse). *Let $d \geq 2$ and $r \geq 1$ be integers. Let P be a full-dimensional d -simplex, let $K \subset P$ be a full-dimensional polytope, and fix a vertex v of P with opposite facet B . Write λ_v for its P -barycentric coordinate and $\rho = 1 - \max_{x \in K} \lambda_v(x)$. If at most $r + 1$ vertices of K lie outside B , some unit direction u parallel to B satisfies*

$$\frac{\pi_P(u) - \pi_K(u)}{\pi_P(u)} \geq \rho^{\min(r, d-1)}. \quad (2.4)$$

Proof. For $\rho = 0$ the result is immediate. Full dimensionality gives $\rho < 1$. Translate v to zero and set $\ell = 1 - \lambda_v$, a linear functional with $\ell(B) = 1$. List the off-base vertices as $q_1, \dots, q_j$, where $1 \leq j \leq r + 1$. Uniquely,

$$q_i = s_i z_i, \quad z_i \in B, \quad \rho \leq s_i < 1.$$

Thus $K \subset \text{conv}(B, q_1, \dots, q_j)$. If $j \geq 2$ and $z_1 \neq z_2$, project orthogonally along $z_1 - z_2$. Their projected base points coincide. Of the projected q_1, q_2 , retain the one with the smaller s_i ; the other lies between it and the common projected base point. If $z_1 = z_2$, the same redundancy exists before projection, and any nonzero base-parallel direction can be chosen. If $j = 1$, choose any such direction and retain this single point.

In every case at most $m \leq r$ inner apex points remain. The functional ℓ descends through the projection. The projected P is a genuine $(d - 1)$ -cone with a $(d - 2)$ -dimensional base; all retained depths are at least ρ . Lemma 2.1 gives missing fraction at least $\rho^{\min(m, d-1)} \geq \rho^{\min(r, d-1)}$. Normalize the projection direction to obtain the required unit u . $\square$

Corollary 2.4 (Actual facet incidence). *Let $d \geq 2$, $r \geq 1$ be integers and $k = \min(r, d - 1)$. Suppose $K \subset P$, where P is a d -simplex and K has at most $d + 1 + r$ vertices. Assume every facet hyperplane of P is the hyperplane of an actual facet of K . If, for $0 \leq \delta \leq 1$,*

$$\pi_P(u) - \pi_K(u) \leq \delta \pi_P(u) \quad (\|u\| = 1),$$

then every P -coordinate satisfies

$$\max_{x \in K} \lambda_i(x) \geq 1 - \delta^{1/k}. \quad (2.5)$$

Proof. An actual facet of a full-dimensional d -polytope contains at least d vertices. Thus at most $r + 1$ vertices of K lie outside any specified facet of P . Apply Lemma 2.3 at each opposite vertex. $\square$

Mere support contact does not give this count: a supporting plane can touch only one vertex. The next two sections construct P from actual facets, rather than assume that a limiting support law supplies this property.

3 A finite determinant assignment theorem

We record a finite version of the retained volume-weighted assignment argument, with its proof. For a real number t , write $t_+ = \max(t, 0)$. All probability measures in this section have finite support, so no integrability or limiting-law issue is present.

Lemma 3.1 (Seminorm assignment). *Let μ be a centered probability measure with finite support $\mathcal{A} \subset \mathbb{R}^d$ whose affine span is $\mathbb{R}^d$. Let N be a seminorm with $N(x) \leq R_0$ on $\mathcal{A}$, where $R_0 \geq 0$. For independent samples with law μ , put*

$$A = \mathbb{E}|\det(X_1, \dots, X_d)|, \quad B = \mathbb{E}|\det((X_0, 1), \dots, (X_d, 1))|.$$

Then $B > 0$, $D := B - A \geq 0$, and there are affinely independent anchors $w_0, \dots, w_d \in \mathcal{A}$ and an assignment $I : \mathcal{A} \rightarrow \{0, \dots, d\}$ with

$$h := \mathbb{E}N(X - w_{I(X)}) \leq R_0(d+1)(d+2)\frac{D}{B}. \quad (3.1)$$

One can choose $I(x)$ to be the least index of a largest barycentric coordinate of x in the anchor simplex.

Proof. For an ordered base $\mathbf{x} = (x_1, \dots, x_d)$ define

$$F_{\mathbf{x}}(y) = \det((x_1, 1), \dots, (x_d, 1), (y, 1)), \quad P_{\mathbf{x}} = \mathbb{E}(F_{\mathbf{x}}(X))_+, \quad M_{\mathbf{x}} = \mathbb{E}(-F_{\mathbf{x}}(X))_+.$$

Centering and affinity in the last column give $|\mathbb{E}F_{\mathbf{x}}(X)| = |\det(x_1, \dots, x_d)|$. Hence

$$D = 2\mathbb{E}_{\mathbf{x}} \min(P_{\mathbf{x}}, M_{\mathbf{x}}) \geq 0. \quad (3.2)$$

For two test points set

$$\psi_{\mathbf{x}}(y, z) = \min((F_{\mathbf{x}}(y))_+, (-F_{\mathbf{x}}(z))_+) + \min((-F_{\mathbf{x}}(y))_+, (F_{\mathbf{x}}(z))_+).$$

Each conditional expected minimum is at most $\min(P_{\mathbf{x}}, M_{\mathbf{x}})$. Therefore

$$\mathbb{E}_{\mathbf{x}, Y, Z} \psi_{\mathbf{x}}(Y, Z) \leq D. \quad (3.3)$$

These statements include singular bases.

For an anchor tuple $W = (w_0, \dots, w_d)$, put $v(W) = |\det((w_0, 1), \dots, (w_d, 1))|$. Form $G(W, x)$ by adding the following witnesses, with any fixed order on each base:

- (i) for each i , the base is the other d anchors and the test points are w_i, x ;
- (ii) for each $i < j$, the base is x and the other $d - 1$ anchors, and the test points are w_i, w_j .

Reversing base orientation does not change ψ . There are $M_d = (d+1)(d+2)/2$ summands. Each uses $d+2$ independent samples when W and x are sampled, so, with $J(W) = \mathbb{E}_X G(W, X)$,

$$\mathbb{E}_W J(W) \leq M_d D, \quad \mathbb{E}_W v(W) = B > 0.$$

Positivity of B follows from an affinely independent tuple in the support. Weighting anchor tuples by $v(W)/B$ gives an anchor tuple with $v(W) > 0$ and

$$\frac{J(W)}{v(W)} \leq M_d \frac{D}{B}. \quad (3.4)$$

Indeed its weighted mean is $\mathbb{E}[J(W)\mathbf{1}_{\{v(W)>0\}}]/B \leq M_d D/B$. No assertion that J vanishes at singular tuples is needed.

Fix this tuple, and write $\alpha_i(x)$ for its barycentric coordinates. Determinant affinity gives, for witnesses of type (i), $v(W) \min(1, (-\alpha_i)_+)$; those of type (ii) equal

$$v(W) [\min((\alpha_i)_+, (\alpha_j)_+) + \min((-\alpha_i)_+, (-\alpha_j)_+)].$$

It follows that $G(W, x) \geq v(W)\phi(x)$, where

$$\phi(x) = \sum_i \min(1, (-\alpha_i(x))_+) + \sum_{i < j} \min((\alpha_i(x))_+, (\alpha_j(x))_+).$$

Let $j = I(x)$ be a largest coordinate. Since $\sum_i \alpha_i = 1$, we have $\alpha_j > 0$. Put $N_x = \sum_i (-\alpha_i)_+$ and $U_x = \sum_{i \neq j} (\alpha_i)_+$. The support bounds yield both

$$N(x - w_j) \leq 2R_0(N_x + U_x), \quad N(x - w_j) \leq 2R_0.$$

The first follows by expanding $x - w_j = \sum_{i \neq j} \alpha_i(w_i - w_j)$; the second uses $x, w_j \in \mathcal{A}$. The pairs containing the largest coordinate give $U_x \leq \sum_{i < l} \min((\alpha_i)_+, (\alpha_l)_+)$. Also $\min(1, N_x) \leq \sum_i \min(1, (-\alpha_i)_+)$. Thus

$$N(x - w_{I(x)}) \leq 2R_0 \min(1, N_x + U_x) \leq 2R_0 \phi(x) \leq 2R_0 \frac{G(W, x)}{v(W)}.$$

Integrating and using (3.4) proves (3.1). The reasoning includes $D = 0$ without division by D . $\square$

4 From the actual facet law to an enclosing simplex

4.1 Raw facet identities and the original defect

For a full-dimensional p -polytope L , let $u_i = s_i n_i$ be its facet area-normal vectors and $b_i = s_i h_L(n_i)$, where n_i is the outward unit normal and $s_i > 0$ is the facet area. Set

$$\mathcal{H}(L) = \sum_{|J|=p} |\det(u_j : j \in J)|, \quad \mathcal{L}(L) = \sum_{|J|=p+1} |\det((u_j, b_j) : j \in J)|. \quad (4.1)$$

The order chosen within a subset is immaterial. Translating L replaces b_i by $b_i - \langle u_i, z \rangle$ and leaves these sums unchanged.

Cauchy's formula gives $b_L(z) = \frac{1}{2} \sum_i |\langle z, u_i \rangle|$. Hence $\Pi L = \sum_i [-u_i/2, u_i/2]$, and the usual volume formula for a zonotope gives $|\Pi L| = \mathcal{H}(L)$. Direct facet parametrization of the unit-height pyramid gives its side pairs and bottom pair as

$$\left(\frac{(u_i, b_i)}{p}, \frac{b_i}{p} \right), \quad ((0, -|L|), 0), \quad (4.2)$$

respectively. One way to check the factor $1/p$ is to integrate the $(p-1)$ -dimensional slices of each side face: its oriented area vector is $(u_i, b_i) \int_0^1 (1-t)^{p-1} dt = (u_i, b_i)/p$. Expanding the corresponding minors gives

$$\mathcal{H}(\text{pyr } L) = p^{-(p+1)} \mathcal{L}(L) + |L| p^{-p} \mathcal{H}(L), \quad (4.3)$$

$$\mathcal{L}(\text{pyr } L) = |L| p^{-(p+1)} \mathcal{L}(L), \quad |\text{pyr } L| = \frac{|L|}{p+1}. \quad (4.4)$$

For the second identity, an all-side lifted minor is zero because its last two coordinate rows coincide; minors using the bottom give the stated expression. Substituting (4.3) into the definition of a therefore proves the exact identity

$$a(L) = \frac{\mathcal{L}(L)}{p|L|\mathcal{H}(L)}. \quad (4.5)$$

These computations concern the original projection bodies, not a substitute invariant. The standard projection covariance $\Pi(AL) = |\det A| A^{-T} \Pi L$ shows that R , and hence a and e , are affine invariant; affine maps of a base extend to its pyramid. The quantity E is affine invariant as well.

4.2 Normalization and the finite cone law

Affinely normalize the prescribed maximum S so that its own centroid is zero and its image contains the closed unit ball. Continue to write K, S for the images. With $n = d+1$, maximum-volume replacement gives

$$|\alpha_i^S(x)| \leq 1 \quad (x \in K). \quad (4.6)$$

Indeed replacing vertex i by x multiplies the absolute volume by $|\alpha_i^S(x)|$. Since $\sum_i \alpha_i^S(x) = 1$ and the centroid is zero, $-x/d$ has S -coordinates $(1 - \alpha_i^S(x))/d \geq 0$. Thus

$$-K \subset dS \subset dK, \quad E(K, S) \leq n. \quad (4.7)$$

The second conclusion uses the exact centered-dilation description

$$g_S + (1+\tau)(S - g_S) = \{x : \alpha_i^S(x) \geq -\tau/n \text{ for all } i\}. \quad (4.8)$$

For the actual irredundant facets F of K , use the notation $n_F, s_F, h_F = h_K(n_F)$ and put

$$X_F = \frac{n_F}{h_F}, \quad p_F = \frac{s_F h_F}{d|K|}, \quad \mu = \sum_F p_F \delta_{X_F}. \quad (4.9)$$

Here $h_F \geq 1$ because the unit ball is in K . The cone-volume identity $\sum_F s_F h_F = d|K|$ and the surface-area balance $\sum_F s_F n_F = 0$ show that μ is a centered probability law. Its atoms are precisely actual facet normals divided by their positive support numbers, with $\|X_F\| \leq 1$ and $h_K(X_F) = 1$. Boundedness of K forces these atoms to span $\mathbb{R}^d$; centeredness with positive weights then gives full affine span. In particular the horizontal and lifted moments A, B of Section 3 are positive.

Cauchy's formula and centeredness give the exact brightness identities

$$m_K(z) := \mathbb{E}(\langle z, X \rangle)_+ = \frac{b_K(z)}{d|K|}, \quad f_K(z) := \mathbb{E}|\langle z, X \rangle| = 2m_K(z). \quad (4.10)$$

Expanding the iid determinants by distinct facet subsets gives

$$A = \frac{d! \mathcal{H}(K)}{(d|K|)^d}, \quad B = \frac{(d+1)! \mathcal{L}(K)}{(d|K|)^{d+1}}.$$

Together with (4.5), this proves

$$a(K) = \frac{B}{nA}, \quad \frac{B-A}{B} = \frac{ne(K)}{1+ne(K)}. \quad (4.11)$$

Jensen's inequality, or (3.2), proves $B \geq A$ and hence $e(K) \geq 0$.

Let $N = h_{K-K}$, a continuous seminorm. At every atom,

$$N(X_F) = h_K(X_F) + h_K(-X_F) \leq 1 + d = n$$

by (4.7). Lemma 3.1, with $R_0 = n$, gives $d+1$ affinely independent *actual atoms* w_i and an assignment with

$$h = \mathbb{E}N(X - w_{I(X)}) \leq \frac{Ce(K)}{1+ne(K)}, \quad C = n^3(n+1). \quad (4.12)$$

Thus every selected w_i is an actual facet normal, a fact that is stronger than merely knowing that an anchor belongs to an unspecified support law.

4.3 Width and same-anchor enclosure

For every $z, y \in \mathbb{R}^d$ one has

$$|\langle z, y \rangle| \leq d m_K(z) N(y). \quad (4.13)$$

To see this, first take unit v and $c = \langle y, v \rangle \neq 0$. In coordinates $x = q + tv$ with $q \in v^\perp$, the determinant-one shear $(q, t) \mapsto (q, t + \langle y, q \rangle / c)$ places K in a cylinder with base the projection of K and height $N(y)/|c|$. Thus $|c||K| \leq N(y)\pi_K(v)$. Scaling v to z and using (4.10) proves (4.13); zero cases are immediate.

Let $p_i = \mu\{I = i\}$. Positive-part Lipschitz continuity gives

$$\sum_i p_i (\langle z, w_i \rangle)_+ \geq (1 - dh) m_K(z). \quad (4.14)$$

Assume $h < 1/(2d)$. On the unit sphere, m_K has a positive lower bound, since K contains the unit ball. Hence for every unit z some anchor has positive pairing with z , uniformly bounded away from zero. Support separation puts zero in the interior of $\text{conv}(w_0, \dots, w_d)$. Its polar

$$P = \{x : \langle w_i, x \rangle \leq 1 \text{ for } 0 \leq i \leq d\} \quad (4.15)$$

is therefore a genuine d -simplex containing K . Every defining inequality is an actual K -facet inequality. The corresponding full facet of K lies in a facet of P , giving exactly the hypothesis of Corollary 2.4.

For $z \in P$, the left side of (4.14) is at most one; therefore $m_K(z) \leq 2$. Equation (4.13) and support separation then give the intrinsic containment

$$P \subset 2d(K - K). \quad (4.16)$$

4.4 Probability correction and relative projection error

Let $\lambda_i > 0$ be the barycentric coordinates of zero in the anchor simplex. Let q_i be the vertex of P opposite the facet with normal w_i . The anchor-simplex coordinates satisfy

$$\alpha_i^w(x) = \lambda_i(1 - \langle q_i, x \rangle). \quad (4.17)$$

Both sides agree at each anchor. The actual cone probabilities of P satisfy the same mass and mean equations as λ_i and hence equal λ_i . Put $c = \sum_i p_i w_i$. Since $\mathbb{E}X = 0$, (4.16) gives $|\langle q_i, c \rangle| \leq 2dh$. Evaluating (4.17) at c yields

$$p_i = \lambda_i(1 - \langle q_i, c \rangle) \geq (1 - 2dh)\lambda_i.$$

Also (4.13) and the absolute-value triangle inequality give

$$\sum_i p_i |\langle z, w_i \rangle| \leq f_K(z) + dm_K(z)h = (1 + dh/2)f_K(z).$$

Write $\theta = 2dh < 1$. The exact brightness formulas for P and K imply

$$(1 - \theta) \frac{\pi_P(u)}{|P|} \leq (1 + \theta/4) \frac{\pi_K(u)}{|K|} \quad (\|u\| = 1). \quad (4.18)$$

The finite first mixed-volume formula and Minkowski's first inequality give

$$\left(\frac{|P|}{|K|} \right)^{1/d} \leq \frac{V(K[d-1], P)}{|K|} = \mathbb{E}h_P(X). \quad (4.19)$$

Indeed the last expression is $\sum_F s_F h_P(n_F)/(d|K|)$. Since $h_P(w_i) = 1$, subadditivity and (4.16) bound it by $1 + 2dh = 1 + \theta$. Combining (4.18) and (4.19),

$$\frac{\pi_K(u)}{\pi_P(u)} \geq \frac{1 - \theta}{(1 + \theta)^d(1 + \theta/4)}.$$

For factors in $[0, 1]$, the deficit of their product is at most the sum of their deficits. Thus the missing fraction above is at most

$$\theta + d\theta + \theta/4 = 2d(d + 5/4)h \leq 3d^2h \quad (d \geq 3). \quad (4.20)$$

Here $1 - (1 + \theta)^{-d} \leq d\theta$ follows from $1 - t^d \leq d(1 - t)$ for $0 \leq t \leq 1$.

Proposition 4.1 (Finite actual-facet enclosure). *For $d \geq 3$, if $0 \leq e(K) \leq H_d^{-1}$, the normalized K is contained in a simplex P whose facet hyperplanes are actual K -facet hyperplanes, and*

$$\pi_P(u) - \pi_K(u) \leq H_d e(K) \pi_P(u) \quad (\|u\| = 1). \quad (4.21)$$

Proof. Since $H_d = 3d^2C$, (4.12) gives $h \leq 1/(3d^2) < 1/(2d)$ even at $e(K) = H_d^{-1}$. All strict enclosure gates hold. Equations (4.20) and (4.12) give $3d^2h \leq H_d e(K)$. $\square$

Every step used the same finite actual facet law. No compact approximation, subsequence, or limiting-law identification is needed for this proposition.

5 Retention of every prescribed maximum simplex

This section reproduces the retained retention estimates with the quantifier on the original simplex explicit. The arguments use only containment and near-vertex points, independently of how those points were obtained.

Lemma 5.1 (Substochastic determinant matching). *Let U be a nonnegative $l \times l$ column-substochastic matrix, where $l \geq 1$. If $|\det U| \geq 1 - \eta$ and $0 \leq \eta < 1/3$, then a permutation of its columns makes every diagonal entry at least $1 - \eta$.*

Proof. Represent each column as the expectation of a random standard basis vector, assigning any missing mass to the zero vector. Independent selection represents the determinant as the expectation of a variable in $\{-1, 0, 1\}$. If every column but column j is sampled, each resulting cofactor vector either vanishes or selects exactly one entry of column j with a sign. Hence $|\det U| \leq \max_i U_{ij}$. Choose an entry at least $1 - \eta$ in each column. If two chosen entries share a row, the independent selections collide with probability at least $(1 - \eta)^2$, so

$$|\det U| \leq 1 - (1 - \eta)^2 \leq 2\eta < 1 - \eta,$$

a contradiction. The selected rows are distinct. $\square$

Lemma 5.2 (Inverse bootstrap). *Let $K \subset P$, where P is a d -simplex and S is a prescribed maximum-volume simplex in K . Suppose $q_i \in K$ have P -coordinates $Q_{ii} \geq 1 - \rho$. Suppose that after relabeling the vertices of this same S , their P -coordinate matrix W satisfies $W_{ii} \geq 1 - \eta$, where $0 \leq \eta \leq 1/8$ and $0 \leq \rho \leq 1/16$. Then*

$$E(K, S) \leq 8(d + 1)\rho. \quad (5.1)$$

Proof. The matrix W is nonnegative column stochastic and invertible. Put $B = W^{-1}$, $D = B - I$, $\beta = \max_{ij} |D_{ij}|$, and $r_j = 1 - W_{jj}$. From $DW = I - W$ and column stochasticity,

$$D_{ij} = (\delta_{ij} - W_{ij}) + \sum_{l \neq j} (D_{ij} - D_{il})W_{lj}.$$

Consequently $\beta \leq \eta + 2\eta\beta$, whence

$$\beta \leq \frac{\eta}{1 - 2\eta} \leq \frac{1}{6}. \quad (5.2)$$

The ii entry of $BW = I$ gives

$$B_{ii} - 1 \geq \frac{(1 - \beta)r_i}{1 - r_i} \geq r_i/2.$$

On the other hand, $q_i \in K$ and maximality of the same S imply $(BQ)_{ii} \leq 1$ by replacement. Since $Q_{ii} \geq 1 - \rho$, $B_{ii} > 0$, and $B_{ij} \geq -\beta$ for $i \neq j$,

$$B_{ii} - 1 \leq \rho(B_{ii} + \beta) \leq 2\rho.$$

Thus $r_i \leq 4\rho$ for every i . Reapply the inverse identity with this improved column loss to obtain

$$\beta \leq \frac{4\rho}{1 - 8\rho} \leq 8\rho.$$

Every S -coordinate of each P -vertex is an entry of B and is at least -8ρ . Convex averaging gives the same lower bound throughout P . Equation (4.8) now yields (5.1) about S 's own centroid. $\square$

Proposition 5.3 (Two all-scale retention bounds). *Let $d \geq 1$ be an integer, $n = d + 1$, and let $K \subset P$, where K is a full-dimensional compact convex body and P a d -simplex. Suppose every P -coordinate attains at least $1 - \rho$ on K , where $\rho \geq 0$. For every prescribed maximum-volume inscribed simplex S ,*

$$E(K, S) \leq 16n^2\rho. \quad (5.3)$$

If, additionally, K is a polytope with at most $n + r$ vertices for an integer $r \geq 1$, then

$$E(K, S) \leq 64nr\rho. \quad (5.4)$$

The local estimate $E(K, S) \leq 8n\rho$ holds in the first case for $\rho \leq 1/(16n)$ and in the second case for $\rho \leq 1/(64r)$.

Proof. The universal estimate $E(K, S) \leq n$ follows from maximum-volume replacement and (4.8), without any normalization. It handles the complementary large- ρ branches of both bounds.

For the unrestricted local branch, choose $q_i \in K$ with P -coordinate at least $1 - \rho$ and let Q be their column coordinate matrix. Independent column selection gives

$$\det Q \geq 2 \prod_i Q_{ii} - 1 \geq 1 - 2n\rho.$$

Indeed the identity selection has probability $\prod_i Q_{ii}$ and every other signed selection is at least -1 . For $\rho \leq 1/(16n)$ the right side is positive, so these points form a comparison simplex. Maximality of S gives $|\det W| \geq \det Q \geq 1 - \eta$ with $\eta = 2n\rho \leq 1/8$. Lemma 5.1 relabels only the vertices of this same S so that $W_{ii} \geq 1 - \eta$. Lemma 5.2 proves the local bound. Together with $E \leq n$ for $\rho > 1/(16n)$, this proves (5.3).

For the bounded-vertex branch assume $\rho \leq 1/(64r)$. Choose actual vertices q_i maximizing the P -coordinates. They are distinct because $\rho < 1/2$. Their nonnegative column-stochastic matrix satisfies

$$\|Q - I\|_1 \leq 2\rho, \quad \|Q^{-1}\|_1 \leq (1 - 2\rho)^{-1} \leq 1 + 4\rho. \quad (5.5)$$

Here $\|\cdot\|_1$ is the induced column-sum matrix norm and $\rho \leq 1/4$. Thus $T = \text{conv}(q_i)$ is a simplex in K .

First let S_v be any maximum simplex with vertices among those of K . Such a maximum exists by multiaffinity of the determinant. It omits at most r of the q_i because there are at most r other vertices. Let R be the omitted index set, of size $l \leq \min(r, n)$. If $l = 0$, then $S_v = T$. Otherwise label its retained q_i by their own indices and put the remaining vertices in columns R . If W is its coordinate matrix and $C_0 = Q^{-1}W$, then $(C_0)_i = e_i$ for $i \notin R$, so

$$|\det(C_0)_{R,R}| = |\det C_0| = \frac{|S_v|}{|T|} \geq 1.$$

The restricted matrix $U = W_{R,R}$ is column substochastic. Each column of $(C_0)_{R,R}$ has ℓ^1 norm at most $1 + 4\rho$; by $W - C_0 = (Q - I)C_0$ its distance from the corresponding U -column is at most $2\rho/(1 - 2\rho) \leq 4\rho$. Telescoping the determinant gives

$$|\det U - \det(C_0)_{R,R}| \leq 4l\rho(1 + 4\rho)^{l-1} \leq 8l\rho.$$

For the last inequality, $4l\rho \leq 1/16$ and $(1 + 4\rho)^{l-1} \leq (1 - 4l\rho)^{-1} < 2$, by the binomial expansion. Hence $|\det U| \geq 1 - \eta$ with $\eta = 8r\rho \leq 1/8$. Lemma 5.1 matches the omitted indices, while the retained q_i already have diagonal coordinates at least $1 - \rho \geq 1 - \eta$. Every maximum vertex simplex therefore has exactly one vertex in each of the disjoint convex caps

$$\mathcal{C}_i = \{x \in K : \lambda_i^P(x) \geq 1 - \eta\}. \quad (5.6)$$

This conclusion also covers $l = 0$.

Now return to the prescribed maximum S , whose vertices need not be vertices of K . Express each vertex of S as a convex combination of K -vertices and sample the combinations independently. Its determinant is the expectation of the sampled determinants. Every sampled absolute volume is at most $|S|$. Equality in the expectation and triangle inequality forces every positive-probability sampled tuple to be a maximum vertex simplex, with one point in each cap (5.6). For each i , some sampled column must lie in $\mathcal{C}_i$ with probability one; otherwise independence would give positive probability that all columns avoid it. Distinct caps require distinct such columns. There are n columns and n caps, so after relabeling, each actual vertex of this same S lies in its corresponding cap by convexity.

Thus $W_{ii} \geq 1 - \eta$ for the prescribed S , and Lemma 5.2 proves $E(K, S) \leq 8n\rho$ locally. For $\rho > 1/(64r)$ the universal estimate gives $E(K, S) < 64nr\rho$. No change of simplex carrier or centroid has occurred. $\square$

6 Assembly of the upper estimate

We prove (1.3), leaving the sharpness statement to the next section. Fix any of the maximum simplices S specified in Theorem 1.1, and normalize that same simplex as in Section 4. The affine map preserves vertex count, maximality, $e(K)$, $E(K, S)$, and the correspondence of centroids.

Suppose first that $0 \leq e(K) \leq H_d^{-1}$. Proposition 4.1 provides an actual-facet-supported simplex P with relative projection error $\delta = H_d e(K) \in [0, 1]$. Since K has at most $n + r$ vertices, Corollary 2.4 gives one point of K in each cap of coordinate loss at most

$$\rho = \delta^{1/k}, \quad k = \min(r, d - 1).$$

Both conclusions of Proposition 5.3 concern the same K, S, ρ . Taking their minimum yields

$$E(K, S) \leq \min(64nr, 16n^2)\rho = T_{d,r} H_d^{1/k} e(K)^{1/k}.$$

At $e(K) = 0$, this gives $\rho = 0$ and $E(K, S) = 0$ without division by any vanishing parameter. At $e(K) = H_d^{-1}$ the strict enclosure gate remains strict, as checked in Proposition 4.1.

For $e(K) > H_d^{-1}$, we have $(H_d e(K))^{1/k} > 1$ and $T_{d,r} \geq n$. Therefore the universal replacement bound $E(K, S) \leq n$ proves the same estimate. Undoing the affine map yields (1.3) for the originally prescribed maximum and its original centroid.

The exponent is supplied by the finite apex count; the constants $64nr$ and $16n^2$ come from retention. Their minimum does not select different maxima or different centers. No classification of all polytopes with two extra vertices, stronger all-minor stochastic conjecture, or forest conjecture enters this argument.

7 The exact truncation and pyramid obstruction

Put $k = \min(r, d - 1)$ and $m = k + 1$, so $2 \leq m \leq d$. For $0 < t < 1$ define

$$C_t = \left\{ x \in \mathbb{R}^m : x_i \geq 0, \quad t \leq \sum_i x_i \leq 1 \right\} = \text{conv}\{e_1, \dots, e_m, te_1, \dots, te_m\}. \quad (7.1)$$

This is a simplex truncated parallel to the facet opposite zero.

Proposition 7.1 (Exact core calculation). *The body C_t has $2m$ vertices, and $T_t = \text{conv}(e_1, \dots, e_m, te_1)$ is a maximum-volume inscribed simplex. With the original centroid of T_t ,*

$$|T_t| = \frac{1-t}{m!}, \quad E(C_t, T_t) = (m+1)t. \quad (7.2)$$

Its actual defect is

$$e_m(C_t) = \frac{t^{m-1}[m(m-1) - (m+1)(m-2)t - 2t^m]}{(m+1)(1-t^m)[m+1 + (m-1)t^{m-1}]} \quad (7.3)$$

In particular,

$$\lim_{t \downarrow 0} \frac{e_m(C_t)}{t^{m-1}} = \frac{m(m-1)}{(m+1)^2} > 0. \quad (7.4)$$

Proof. Every full-dimensional vertex simplex must use all m coordinate rays; otherwise it lies in a coordinate hyperplane. An $(m+1)$ -tuple consequently has exactly one doubled ray. Its determinant magnitude is $(1-t)t^j$ for some $j \geq 0$: the doubled ray contributes $1-t$ and each other selected low vertex contributes a factor t . Hence the maximum vertex-simplex volume

is $(1-t)/m!$, attained by T_t . Expressing arbitrary inscribed points as convex combinations of the vertices and using multiaffinity and the triangle inequality proves the same bound for all inscribed simplices.

Write $s = \sum_i x_i$. Relative to the ordered vertices $(te_1, e_1, e_2, \dots, e_m)$ of T_t , the coordinates are

$$\beta_0 = \frac{1-s}{1-t}, \quad \beta_1 = x_1 - t\beta_0, \quad \beta_i = x_i \quad (2 \leq i \leq m).$$

On C_t , $0 \leq \beta_0 \leq 1$ and all coordinates are at least $-t$. At te_2 , the coordinate β_1 equals $-t$. The centered-dilation formula (4.8) proves (7.2).

For the defect put $c = 1/(m-1)!$, $q = t^{m-1}$, and $b = 1 - q$. The actual facet area-normal and support pairs are

$$(-cbe_i, 0) \quad (1 \leq i \leq m), \quad (c\mathbf{1}, c), \quad (-cq\mathbf{1}, -ct^m).$$

The volume is $(1-t^m)/m!$. The horizontal minors consist of the all-coordinate subset, the m subsets using the top facet, and the m subsets using the bottom facet; subsets using both parallel top and bottom normals vanish. Therefore

$$\mathcal{H}(C_t) = c^m b^{m-1} [m+1 + (m-1)q]. \quad (7.5)$$

For the lifted minors the all-coordinate-plus-top term is $c^{m+1}b^m$, the all-coordinate-plus-bottom term is $c^{m+1}b^m t^m$, and each of the m terms with top, bottom, and $m-1$ coordinate facets is $c^{m+1}b^{m-1}(q - t^m)$. Thus

$$\mathcal{L}(C_t) = c^{m+1}b^{m-1} [1 + (m-1)q - (m-1)t^m - qt^m]. \quad (7.6)$$

By (4.5),

$$a_m(C_t) = \frac{1 + (m-1)q - (m-1)t^m - qt^m}{(1-t^m)[m+1 + (m-1)q]}.$$

Subtracting $1/(m+1)$ gives (7.3). Its numerator bracket equals $(m+1)(m-2)(1-t) + 2(1-t^m) > 0$ for every $m \geq 2$. The limit (7.4) follows. $\square$

Remark 7.2 (The two-dimensional core). The computation explicitly includes $m = 2$; it is not an invocation of a statement restricted to dimensions at least three. In this case

$$e_2(C_t) = \frac{2t}{3(3+t)}, \quad 3 - \frac{1}{a_2(C_t)} = \frac{2t}{1+t}. \quad (7.7)$$

This is the core required for the entire $r = 1$ layer of the main theorem.

Lemma 7.3 (Pyramid identities). *For a full-dimensional p -polytope L with $a_p(L) > 0$,*

$$a_{p+1}(\text{pyr } L) = \frac{a_p(L)}{1 + a_p(L)}. \quad (7.8)$$

Consequently

$$\Delta(L) := p + 1 - \frac{1}{a_p(L)} \quad (7.9)$$

is unchanged by taking a pyramid. If T is a maximum simplex in L , then $\text{pyr } T$ is a maximum simplex in $\text{pyr } L$. If all T -coordinates on L are at least $-t$, and one coordinate attains $-t$, then the same is true for the $\text{pyr } T$ -coordinates on $\text{pyr } L$.

Proof. Substitute (4.3)–(4.4) into (4.5) to get

$$a_{p+1}(\text{pyr } L) = \frac{\mathcal{L}(L)}{\mathcal{L}(L) + p|L|\mathcal{H}(L)} = \frac{a_p(L)}{1 + a_p(L)}.$$

Thus $1/a_{p+1} = 1 + 1/a_p$, proving (7.9). Every full-dimensional vertex simplex in a pyramid includes its unique apex, and its base is a full-dimensional vertex simplex of L . Comparing base volumes proves the maximum statement for vertex simplices; multiaffinity extends it to arbitrary inscribed simplices. A point in the pyramid can be written as $((1-s)x, s)$ with $x \in L$, $0 \leq s \leq 1$. Its old coordinates are $(1-s)\alpha_i^T(x)$ and its new apex coordinate is s . They are bounded below by $-t$, and the old base still realizes that lower value. $\square$

Proof of sharpness in Theorem 1.1. Set

$$K_t = \text{pyr}^{d-m} C_t, \quad S_t = \text{pyr}^{d-m} T_t.$$

These are actual d -polytopes and actual prescribed maxima. Their vertex count is $2m + (d-m) = d+1+k \leq d+1+r$. Iterating Lemma 7.3 and using (7.2) gives

$$E(K_t, S_t) = nt, \quad n = d+1. \quad (7.10)$$

From (7.4) and the identity

$$\Delta(C_t) = \frac{(m+1)^2 e_m(C_t)}{1 + (m+1)e_m(C_t)},$$

we obtain $\Delta(C_t)/t^k \rightarrow k(k+1)$. In ambient dimension d , pyramid invariance and $a_d = 1/(n-\Delta)$ yield exactly

$$e_d(K_t) = \frac{\Delta(C_t)}{n[n-\Delta(C_t)]}, \quad \frac{e_d(K_t)}{t^k} \rightarrow \frac{k(k+1)}{n^2}. \quad (7.11)$$

Together with (7.10), this is (1.4). For $\beta > 1/k$, the ratio $E(K_t, S_t)/e_d(K_t)^\beta$ is asymptotic to a positive constant times $t^{1-k\beta}$, which diverges. Hence no finite coefficient depending on d, r can support that stronger exponent near zero defect. $\square$

8 Exact sharpness of the geometric projection step

The preceding family proves sharpness for the original invariant. Here we separately verify sharpness of the existence-of-direction bound in Lemma 2.3. Equality in just one direction would not suffice for this purpose; an upper bound in every direction is needed.

Fix integers $1 \leq r \leq d-1$, put $m = r+1$, and let $0 < \rho < 1$. Set

$$\begin{aligned} P &= \text{conv}(0, e_1, \dots, e_d), \\ K &= \text{conv}(e_1, \dots, e_d, \rho e_1, \dots, \rho e_m) \\ &= \left\{ x \geq 0 : \sum_i x_i \leq 1, \quad \rho^{-1} \sum_{i \leq m} x_i + \sum_{i > m} x_i \geq 1 \right\}. \end{aligned}$$

The barycentric loss at zero is ρ . This body has $d+m = d+1+r$ vertices and is an iterated pyramid over the m -dimensional equal truncation. Every facet plane of P supports a full facet of K .

With $c = 1/(d-1)!$, the actual facet area-normal vectors of K are

$$\begin{aligned} & -c(1 - \rho^{m-1})e_i && (i \leq m), \\ & -c(1 - \rho^m)e_i && (i > m), \\ & c\mathbf{1}, && -c(\underbrace{\rho^{m-1}, \dots, \rho^{m-1}}_m, \underbrace{\rho^m, \dots, \rho^m}_{d-m}). \end{aligned}$$

For a coordinate facet the removed simplex has the corresponding product of intercept scales; the last vector is obtained from the intercepts of the new cut facet. These vectors sum to zero, as required.

For nonzero $z \in \mathbb{R}^d$ write

$$A = \sum_{i \leq m} z_i, \quad B = \sum_{i > m} z_i, \quad A_* = \sum_{i \leq m} |z_i|, \quad B_* = \sum_{i > m} |z_i|.$$

Cauchy's formula gives the exact *homogeneous* brightness ratio

$$\frac{b_P(z) - b_K(z)}{b_P(z)} = \rho^{m-1} \frac{A_* + \rho B_* - |A + \rho B|}{A_* + B_* + |A + B|}. \quad (8.1)$$

The denominator is positive. The numerator is nonnegative by the triangle inequality and is at most the denominator. Therefore the relative deficit is at most $\rho^{m-1} = \rho^r$ in every direction. For $z = e_1 - e_2$, the ratio equals ρ^r , and that direction is parallel to the opposite base facet of P . Upon normalizing z , we conclude

$$\max_{\|u\|=1} \frac{\pi_P(u) - \pi_K(u)}{\pi_P(u)} = \rho^r. \quad (8.2)$$

The same maximum is attained among base-parallel unit directions. Thus both the exponent and the constant one in Lemma 2.3 are sharp throughout $1 \leq r \leq d-1$. The distinction between $b_K(z)$ and $\pi_K(u)$ avoids any convention for projection notation at nonunit vectors.

A Source correspondence and verification boundary

This manuscript expands the frozen proof `research_math/vertex_excess_sharp_exponent_20261008/SHARP_VERTEX_EXCESS_EXPONENT.md` and its geometric-sharpness addendum. The frozen main has SHA-256 `8da4c012d4099465b4702ce4091ba4d1e2cc7862c12e8eb55cbd7467414e0f05`. The separate traditional review is `research_math/vertex_excess_exponent_independent_audit_20261008/REFeree_REPORT.md`, SHA-256 `d638e1ebf4505f480d99c08c21a2012fb36f0e021cd86d828d142effd5df283b`. The supplied review checks the main exponent, the exact actual-defect constant, the arbitrary-prescribed-maximum quantifier, both defect endpoints, and the lower family. It is an independent model's analytic review, not a human referee report or a new Lean replay.

A.1 Retained arguments reproduced here

The volume-weighted determinant selection is retained from `notes/quadratic-dimensional-simplex-stability/imports/weighted_anchors.md` in the project publication tree. Section 3 supplies its finite-support seminorm proof. The bounded-vertex retention argument is retained from `research_math/linear_retention_investigation_20261008/BOUNDED_VERTEX_EXCESS_RETENTION.md`. Section 5 includes both that proof and the unrestricted retention proof. The exact truncation calculation is retained from `notes/simplex-truncation-stability/proof.tex` in the publication tree; Section 7 includes the facet calculation, the $m = 2$ case, the maximum-simplex proof, and the pyramid identities. The package's `SOURCE_MAP.md` records the precise local source paths, hashes, and section-level correspondence, and includes frozen copies of the main input documents.

A.2 Lean references refer to retained statements only

For the earlier formal source, the module filenames and theorem namespaces must be distinguished. In particular, the retained interfaces include

- `Entry005.finite_halfspace_entryA_lifted_moment`, in `Entry005/PyramidEntryDefect.lean`;
- `Entry005.finite_halfspace_defect_over_first_moment`, in `Entry005/PyramidMomentDefect.lean`;
- `Entry005.unit_ball_seminorm_first_moment_assignment`, in `Entry005/SeminormFirstMoment.lean`.

Actual finite-law probability, centering, ball support, and brightness interfaces are in `Entry005/FiniteHalfspaceConeLaw.lean`; finite-support assignment interfaces are in `Entry005/FiniteBodyWeightedAssignment.lean`; the earlier unrestricted prescribed-simplex interface is in `Entry005/RelativeSimplexRetention.lean`. These are references to retained code, not new formalization claims for this manuscript. In particular, `PyramidEntryDefect` and `PyramidMomentDefect` are module names, not the theorem namespace.

The older compact-limit main theorem does not by itself provide the actual-facet incidence used in Corollary 2.4. Here that incidence is proved directly by choosing anchors from the finite law (4.9). The new cone/prismoid estimate, ray-collapse/counting step, finite-law specialization with this incidence, and final exponent assembly still need corresponding Lean formalization. An earlier successful `compiledMain` or other main build cannot certify those additions.

A.3 Editorial clarifications implemented

The positive-integer restrictions in Lemma 2.1, unit projection convention and homogeneous brightness notation, explicit two-dimensional core, actual Lean namespace references, and mixed-volume factorial convention implement all nonblocking clarifications requested by the supplied review. The finite actual-facet chain is expanded in full in Section 4. Standard mixed-volume inputs remain credited as classical, and no strict-monotonicity theorem from [1] is needed.

The theorem is a mathematical manuscript candidate. Neither the focused literature check nor the separate model review establishes priority, journal suitability, or acceptance by professional mathematicians.

References

- [1] F. Bihan and I. Soprunov, *Criteria for strict monotonicity of the mixed volume of convex polytopes*, arXiv:1702.07676v2 (2017). <https://arxiv.org/abs/1702.07676v2>.
- [2] R. Schneider, *Convex Bodies: The Brunn–Minkowski Theory*, second expanded edition, Encyclopedia of Mathematics and its Applications 151, Cambridge University Press, 2014.
- [3] R. Schneider, *Stability for some extremal properties of the simplex*, Journal of Geometry **96** (2009), 135–148. <https://doi.org/10.1007/s00022-010-0028-0>.
- [4] K. J. Böröczky and M. Henk, *Cone-volume measure and stability*, Advances in Mathematics **306** (2017), 24–50. <https://arxiv.org/abs/1407.7272>.
- [5] C. Saroglou, *Volumes of projection bodies of some classes of convex bodies*, Mathematika **57** (2011), 329–353. <https://doi.org/10.1112/S0025579311001860>.
- [6] N. S. Brannen, *Volumes of projection bodies*, Mathematika **43** (1996), 255–264. <https://doi.org/10.1112/S002557930001175X>.